

Activity 2: Requirement Analysis and Prompt Engineering

Purpose

Requirement analysis is one of the most important phases of the Software Development Life Cycle (SDLC). Before development begins, software engineers analyze requirements, identify functional and non-functional requirements, define business rules, and plan the application's structure. In this activity, you will use Generative AI to perform requirement analysis and learn how different prompting techniques influence the quality of AI-generated responses.

Pre-requisites

- Completion of Activity 1
 - ChatGPT or Oracle Chat AI
 - Basic understanding of software requirements
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Estimated Duration

40 Minutes

Scenario

After understanding the SmartBank project, your Team Lead asks you to assist the Business Analyst in preparing the Software Requirements Specification (SRS).

Instead of manually analyzing the requirements, you will use AI to organize the information into meaningful software artifacts while experimenting with different prompt engineering techniques.

Tasks

Complete the following tasks in sequence.

Task 1: Identify Functional and Non-Functional Requirements

Use AI to distinguish between functional requirements (what the system should do) and non-functional requirements (how the system should perform).

Prompt

Act as an experienced Business Analyst.

Analyze the SmartBank Net Banking Application.

Identify:

- Functional Requirements
- Non-Functional Requirements

For each requirement include:

- Requirement ID
- Description
- Priority (High/Medium/Low)

Present the output as a structured table.

Task 2: Generate User Stories

Convert the business requirements into Agile user stories.

Prompt

Act as an Agile Product Owner.

Generate user stories for the following SmartBank modules:

- Login
- Dashboard
- Fund Transfer
- Beneficiary Management
- Transaction History

For each module provide:

- User Story
- Acceptance Criteria
- Priority

Present the output in a table.

Task 3: Zero-Shot Prompting

Generate requirements without providing any examples.

Prompt

Generate detailed functional requirements for the Fund Transfer module of a Net Banking application.

Observe the response carefully before moving to the next task.

Task 4: Few-Shot Prompting

Now provide AI with an example before asking it to generate similar content.

Prompt

Act as a Business Analyst.

Example

Module: Customer Login

Functional Requirements

- User enters username and password.
- System validates credentials.
- Display dashboard after successful login.
- Display an appropriate error message for invalid credentials.

Using the same format, generate functional requirements for the Fund Transfer module.

Compare this output with the response generated in Task 3.

Task 5: Chain-of-Thought Prompting

Ask AI to reason through the problem before generating the final response.

Prompt

Act as a Senior Solution Architect.

Before generating the functional requirements for the Fund Transfer module, think step by step.

Consider:

- User actions
- Business validations
- Security checks
- Error scenarios
- Successful transaction flow

Finally generate complete functional requirements.

Compare this response with the previous two techniques.

Task 6: Improve a Weak Prompt

The following prompt is too vague.

Create a login page.

Rewrite it into a detailed engineering prompt.

Example Prompt

Act as a UI Solution Designer.

Design a responsive Login page for the SmartBank Net Banking Application.

Include:

- Username
- Password
- Remember Me
- Forgot Password
- Login button
- Validation messages

Describe the layout using sections and bullet points.

Do not generate HTML or CSS.

Compare both outputs.

Task 7: Generate Structured Output

Software teams often require outputs in structured formats.

Prompt

Generate the following information for the SmartBank application in JSON format.

Include:

- Modules
- Features
- User Roles
- REST APIs
- Database Tables

Ensure the JSON is properly formatted.

Expected Deliverables

By the end of this activity, you should have:

- Functional Requirements
 - Non-Functional Requirements
 - User Stories
 - Acceptance Criteria
 - Zero-Shot Prompt Output
 - Few-Shot Prompt Output
 - Chain-of-Thought Prompt Output
 - Improved Prompt Example
 - Structured JSON Documentation
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Challenge Task

Your manager wants to estimate the development effort for the SmartBank application.

Use AI to prepare a high-level project estimation.

Prompt

Act as an Enterprise Project Manager.

Based on the SmartBank Net Banking Application, prepare a high-level project estimation.

Include:

- Major Modules
- Estimated Development Complexity (Low, Medium, High)
- Suggested Team Responsible
- Estimated Development Effort (Small, Medium, Large)

Present the information as a project planning table suitable for project kickoff.